proof of the inequalities

Link between plain complexity and prefix complexity

We want to prove: $C(s) + O(1) \leq K(s) < C(s) + 2 \times \log(C(s)) + O(1)$

The first inequality is a consequence of the invariance theorem, since prefix machines are particular Turing machines.

A minimal program p that generates s is such that $|p| = C(s)$. A self-delimited program can be obtained by coding p in a self-delimited way (*e.g.* by prefixing it by its double-coded length). This double-length coded version of p is self-delimited. Therefore, $K(s) < 2\log(|p|) + |p| + O(1)$, which is the second inequality.

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